

Package ‘Keeper’

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Title An R package to review patient profiles for phenotype validation

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Description An R package to review patient profiles for phenotype validation.

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computeCohortOperatingCharacteristics
<i>Compute cohort operating characteristics based on a Reference Cohort</i>

Description

Computes operating characteristics (sensitivity, specificity, positive predictive value, AUC, and Cohen’s kappa) of a cohort definition by comparing it against a reference cohort created from LLM review of KEEPER profiles. Metrics are computed separately for high-certainty reviews, low-certainty reviews, and all reviews combined.

Usage

```
computeCohortOperatingCharacteristics(  
  connectionDetails = NULL,  
  connection = NULL,  
  cohortDatabaseSchema,  
  cohortTable,  
  cohortDefinitionId,  
  referenceCohortDatabaseSchema,  
  referenceCohortTableNames,  
  referenceCohortDefinitionId,  
  type = "incident",  
  washoutPeriod = 0,  
  stratifyByCertainty = FALSE  
)
```

Arguments

connectionDetails
An R object of type connectionDetails created using the [DatabaseConnector::createConnection](#) function. Not required if connection is provided.

connection	The connection to the database server created using <code>DatabaseConnector::connect()</code> . Not required if connectionDetails is provided.
cohortDatabaseSchema	The name of the database schema containing the cohort to evaluate.
cohortTable	The table name containing the cohort to evaluate.
cohortDefinitionId	The cohort definition ID of the cohort to evaluate.
referenceCohortDatabaseSchema	The name of the database schema containing the reference cohort (as uploaded by <code>uploadReferenceCohort()</code>).
referenceCohortTableNames	The table names where the reference cohort and metadata are stored. Should be created using <code>[createReferenceCohortTableNames()]</code> . <code>[createReferenceCohortTableNames()]</code> : R:createReferenceCohortTableNames())
referenceCohortDefinitionId	The cohort definition ID of the reference cohort.
type	If type = "incident", phenotypes are evaluated to also get the right cohort start date. If type = "prevalent", the evaluation only asks whether the phenotype correctly classified a person as case or non-case.
washoutPeriod	The minimum required continuous observation time prior to index date for a person to be included in the cohort. People with a cohort start date within the washout period will be removed before computing performance metrics.
stratifyByCertainty	Stratify the output by LLM certainty level?

Value

A tibble with one row per certainty level ("high", "low", "all") and columns for true positives, false positives, true negatives, false negatives, sensitivity, specificity, PPV, NPV (each with lower and upper confidence bounds), AUC, kappa, disease prevalence and certainty.

Specificity, NPV, and prevalence are computed both within the reference cohort, and, based on the prevalence of the highly- sensitive cohort, also in the overall population.

convertKeeperToTable	<i>Convert Keeper profiles to a table</i>
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Description

Convert Keeper profiles to a table

Usage

```
convertKeeperToTable(keeper, removePii = FALSE)
```

Arguments

keeper	Keeper profiles as created using the <code>generateKeeper()</code> function.
removePii	Remove person identifiable information such as person ID and cohort start date from the output? patient data later.

Value

A tibble with one row per person, and text columns for each Keeper category. Can be used for manual review.

`createPromptSettings` *Create settings for generating prompts*

Description

Create settings for generating prompts

Usage

```
createPromptSettings(maxDays = 20, legacy = FALSE)
```

Arguments

<code>maxDays</code>	How many days can a single code have? For example, if <code>maxDays = 5</code> and there is a measurement code that appears on more than 5 days, a random sample of 5 days will be taken. Set to 0 if there is no maximum.
<code>legacy</code>	IF TRUE, will use the prompt from the Nature Digital Medicine paper.

Value

A settings object, to be used in `reviewCases()`.

`createReferenceCohortTableNames`
Create reference cohort table names

Description

Derives a metadata table name from the reference cohort table name in a systematic way.

Usage

```
createReferenceCohortTableNames(referenceCohortTable)
```

Arguments

<code>referenceCohortTable</code>	The name of the cohort table itself. The metadata table name will be derived from this by appending <code>'_metadata'</code> .
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Value

A list with `referenceCohortTable` and `referenceCohortMetadataTable`.

`createSensitiveCohort` *Create a highly-sensitive cohort for a phenotype*

Description

Creates a cohort that is highly sensitive. Nobody outside the cohort is expected to have the disease. As a consequence, specificity inside the cohort is likely low.

This cohort can be used to determine the sensitivity of a cohort definition.

The cohort is constructed by combining two sets:

1. Anybody who has one of the condition concepts of the disease of interest, indexed on the first occurrence.
2. Anybody who has 'highly specific concepts' from at least two Keeper categories (symptoms, drugs, etc.), indexed on the first occurrence.

Highly specific concepts are concepts where at least 10% of those who have that concept also have a condition concept for the disease of interest.

Usage

```
createSensitiveCohort(
  connectionDetails = NULL,
  connection = NULL,
  cdmDatabaseSchema,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  cohortDatabaseSchema,
  cohortTable,
  createCohortTable = FALSE,
  cohortDefinitionId,
  keeperConceptSets,
  minDoiPersons = 100
)
```

Arguments

`connectionDetails`

An R object of type `connectionDetails` created using the `DatabaseConnector::createConnectionDetails` function. Not required if connection is provided.

`connection`

The connection to the database server created using `DatabaseConnector::connect()`. Not required if `connectionDetails` is provided.

`cdmDatabaseSchema`

The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example `'cdm_instance.dbo'`.

`tempEmulationSchema`

Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

`cohortDatabaseSchema`

The name of the database schema that is the location where the cohort will be stored.

cohortTable	The table name where the cohort will be stored.
createCohortTable	Create the cohort table. If TRUE and the table already exists it will first be deleted.
cohortDefinitionId	The cohort definition ID that will be used for the cohort.
keeperConceptSets	A data frame with Keeper concept sets as generated by generateKeeperConceptSets() .
minDoiPersons	Minimum number of persons with DoI concepts. If the count is lower and the keeperConceptSets has a 'hypernym' category that is used instead. Else, an error is thrown.

Value

This function is primarily executed for instantiating the cohort in the cohort table. It does return a data frame listing the concepts that were used to create the cohort.

generateKeeper	<i>Generate Keeper profiles</i>
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Description

Generates Keeper profile for (a subset of) persons in the provided cohort.

Usage

```
generateKeeper(
  connectionDetails = NULL,
  connection = NULL,
  cdmDatabaseSchema,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  cohortDatabaseSchema,
  cohortTable,
  cohortDefinitionId,
  sampleSize = 20,
  personIds = NULL,
  keeperConceptSets,
  phenotypeName = NULL,
  useDescendants = TRUE,
  removePii = TRUE
)
```

Arguments

connectionDetails	An R object of type connectionDetails created using the DatabaseConnector::createConnectionDetails() function. Not required if connection is provided.
connection	The connection to the database server created using DatabaseConnector::connect() . Not required if connectionDetails is provided.

cdmDatabaseSchema	The name of the database schema that contains the OMOP CDM instance. Requires read permissions to this database. On SQL Server, this should specify both the database and the schema, so for example 'cdm_instance.dbo'.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
cohortDatabaseSchema	The name of the database schema that is the location where the cohort to review is stored.
cohortTable	The tablename that contains the cohort to review.
cohortDefinitionId	The cohort definition ID that will be used for the cohort.
sampleSize	The maximum number of persons to randomly sample from the cohort.
personIds	Only persons with these IDs will be selected. Can be a vector of numeric or string (to avoid integer overflows).
keeperConceptSets	A data frame with Keeper concept sets as generated by generateKeeperConceptSets() .
phenotypeName	(Optional) Name of the phenotype. Will be included in the output.
useDescendants	Include the descendants of the concepts specified in the keeperConceptSets, or use only the verbatim concepts?
removePii	Remove person identifiable information such as person ID and cohort start date from the output? Can be set to FALSE to allow the Keeper results to be joined to patient data later.

Value

Returns a tibble with fields

- generatedId: A person-specific ID generated by this function.
- startDay
- endDay
- conceptId
- conceptName
- category: "symptoms", "priorDisease", ...
- target: Can be "Disease of interest", "Alternative diagnoses", "Both", "Other", or NA

This contains both the Keeper profiles themselves, as well as some meta-data such as the CDM source name, and cohort prevalence.

generateKeeperConceptSets

Generate Keeper input concept sets

Description

Generate the concept sets used as input for Keeper. This function uses LLMs, the OHDSI vocabulary vector store, and Phoebe to populate concept sets for the disease of interest, symptoms, treatments etc. of the given medical condition.

Usage

```
generateKeeperConceptSets(
  phenotype,
  clinicalDefinition = NULL,
  client,
  vocabConnectionDetails,
  vocabDatabaseSchema
)
```

Arguments

phenotype	A text string denoting the condition of interest, e.g. 'Type I Diabetes Mellitus (T1DM)'. This string is used as input for the LLM.
clinicalDefinition	Optionally provide a text blob with the definition and any other information about the phenotype.
client	An LLM client created using the <code>ellmer</code> package.
vocabConnectionDetails	Connection details for a database server hosting the OHDSI Vocabulary. Should be created using the createConnectionDetails function in the <code>DatabaseConnector</code> package.
vocabDatabaseSchema	The name of the database schema on the server where the vocabulary tables are located.

Value

A data frame with the following columns:

- conceptId
- conceptName
- vocabularyId
- conceptSetName: 'doi', 'symptoms', ...
- target: 'Disease of interest' or 'Alternative diagnoses'.

launchReviewerApp	<i>Launch the reviewer Shiny app</i>
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Description

Launch the reviewer Shiny app

Usage

```
launchReviewerApp(keeper, keeperConceptSets, decisionsFileName)
```

Arguments

keeper	The output of the generateKeeper() function.
keeperConceptSets	The output of the generateKeeperConceptSets() function.
decisionsFileName	The location of the CSV file where the decisions made by the reviewer will be written. If the file doesn't exist it will be created. If it does exist, it will be loaded and can be edited in the Shiny app.

Value

Returns nothing. Called for launching the Shiny app.

reviewCases	<i>Review Keeper profiles using an LLM</i>
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Description

Review Keeper profiles using an LLM

Usage

```
reviewCases(  
  keeper,  
  settings = createPromptSettings(),  
  phenotypeName = NULL,  
  clinicalDefinition = NULL,  
  client,  
  cacheFolder  
)
```

Arguments

keeper	Output from the generateKeeper() function.
settings	Prompt creating settings as created using the createPromptSettings function.
phenotypeName	The name of the disease to use in the prompt. If not provided, the name in the Keeper input will be used.
clinicalDefinition	Optionally provide a text blob with the definition and any other information about the phenotype.
client	An LLM client created using the <code>ellmer</code> package.
cacheFolder	A folder where the LLM responses are cached. If the process terminates for some reason, it can pick up where it left off using the cache.

Value

A tibble with these columns:

- `generatedId`
- `isCase`, with possible values "yes" or "no",
- `certainty`, certainty of the LLM in its decision, can be "high" or "low".
- `justification`, written by the LLM.
- `cohortPrevalence`, prevalence of the cohort in the entire population.
- `model`, the LLM used to review.
- `keeperVersion`, the version of the Keeper package.

When the Keeper profiles were generated with `removePii = FALSE`, the following columns are also included:

- `personId`
- `cohortStartDate`

uploadReferenceCohort *Upload a reference cohort*

Description

A reference cohort is typically a large sample (e.g. 10,000 persons) of a highly-sensitive cohort (as created using [createSensitiveCohort\(\)](#)), reviewed by an LLM (using [reviewCases\(\)](#)).

The reference cohort can be used to compute operating characteristics of a cohort definition for the same phenotype using [computeCohortOperatingCharacteristics\(\)](#).

Usage

```
uploadReferenceCohort(
  connectionDetails = NULL,
  connection = NULL,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  referenceCohortDatabaseSchema,
  referenceCohortTableNames,
  referenceCohortDefinitionId,
  createReferenceCohortTables = FALSE,
  reviews
)
```

Arguments

connectionDetails	An R object of type connectionDetails created using the DatabaseConnector::createConnectionDetails() function. Not required if connection is provided.
connection	The connection to the database server created using DatabaseConnector::connect() . Not required if connectionDetails is provided.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
referenceCohortDatabaseSchema	The name of the database schema where the reference cohort will be stored.
referenceCohortTableNames	The table names where the reference cohort and metadata will be stored. Should be created using createReferenceCohortTableNames() .
referenceCohortDefinitionId	The cohort definition ID that will be used for the reference cohort.
createReferenceCohortTables	Create the reference cohort and metadata tables? If TRUE and the tables already exists they will first be deleted.
reviews	A data frame containing reviews as generated by reviewCases() .

Value

This function does not return a value. It is called for the side effect of uploading the reference cohort to the database.

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